Assignment 3 – Cyber Security
Linkchain Gaming
Task 1 – Security Risk Assessment

(I have been asked to undertake a cyber security risk assessment for Linkchain Gaming, a medium-sized UK game development company. Following a recent cyber security incident involving a disgruntled employee sharing login credentials, the company has identified the need to strengthen security controls and modernise the enterprise network.

This report identifies threats, vulnerabilities, assets at risk, likelihood, impact and recommended controls. The aim is to reduce the likelihood of future incidents, protect confidential intellectual property, secure remote workers, and improve business resilience.

Risk Assessment Table

	Threat	**Vulnerability**	**Asset**	**Impact**	**Likelihood**	**Risk**	**Recommended Action**	**Control Type**
	Unauthorised access using shared credentials	Staff shared username/password	Confidential files	High	High	High	Enforce unique accounts and MFA	Preventative
	Data theft by insider	Excessive shared folder permissions	Intellectual Property	High	High	High	Apply RBAC and least privilege	Preventative

3	Malware infection	Jordan opens all attachments automatically	End user devices	High	High	High	Email filtering and awareness training	Preventative
4	Phishing attack	Lack of user awareness	Staff accounts	High	High	High	Phishing simulation and training	Preventative
5	Ransomware	No mention of secure backups	Servers / data	High	Medium	High	Daily cloud backups and offline copies	Corrective
6	Remote compromise	Home working unsecured network	Remote devices	High	Medium	High	VPN with MFA and device compliance checks	Preventative
7	Application crashes	Outdated hardware/software	Workstations	Medium	High	Medium	Upgrade RAM/SSD and patch software	Corrective
8	Loss of work	No autosave or backup sync	User files	Medium	High	Medium	Enable OneDrive autosave/versioning	Corrective

	Credential reuse	Weak password practices	User accounts	High	High	High	Password policy + MFA	Preventative
	USB malware	Removable media unrestricted	PCs / network	Medium	Medium	Medium	Block unknown USB devices	Preventative
	Lack of monitoring	No SIEM / alerting	Entire network	High	Medium	High	Implement logging and SIEM	Detective
	Privilege abuse	Junior users may have admin rights	Endpoints	High	Medium	High	Remove local admin rights	Preventative
	Denial of Service	No ISP redundancy	Internet services	High	Medium	High	Dual ISP failover	Corrective
	Firewall gaps	Legacy perimeter controls	Network	High	Medium	High	Next-gen firewall	Preventative
	Patch vulnerabilities	Missing updates	PCs / servers	High	High	High	Central patch management	Preventative
	Unencrypted	Laptop theft risk	Remote laptops	High	Medium	High	BitLocker full disk	Preventative

devices						encryption	
No asset register	Devices unmanaged	Hardware	Medium	Medium	Medium	Create CMDB asset register	Administrative
Third-party compromise	Supplier access unmanaged	Systems	Medium	Medium	Medium	Supplier security reviews	Administrative
Poor incident response	No tested procedure	Operations	High	Medium	High	Create IR plan and run exercises	Corrective
Reputational damage	IP leaks online	Company brand	High	Medium	High	DLP + legal + monitoring	Corrective

Priority Remediation Actions

Immediate (0–30 Days)

1. Enable Multi-Factor Authentication for all users
2. Reset all passwords after previous credential sharing incident
3. Remove unnecessary shared folder access
4. Scan all devices for malware
5. Deploy advanced email filtering
6. Patch all endpoints and servers

Short Term (1–3 Months)

1. Implement Role Based Access Control
2. Deploy SIEM logging solution
3. Introduce VPN for remote workers

4. Create backup strategy (3-2-1 model)
5. Replace slow/outdated hardware causing crashes

Long Term (3–12 Months)

1. ISO 27001 aligned ISMS
2. Disaster Recovery environment
3. Cyber Essentials Plus certification
4. Annual penetration testing
5. Security awareness programme

Physical Controls

- Secure office entry with access cards
- CCTV monitoring
- Locked server rooms
- Visitor sign-in system
- Screen lock policy

These reduce unauthorised physical access to systems and confidential development material.

Administrative Controls

- Acceptable Use Policy
- Password Policy
- Joiners / Movers / Leavers process
- Incident Response Policy
- Data Classification Policy
- Mandatory cyber awareness training

These controls ensure staff understand responsibilities and reduce human error.

Technical Controls

- MFA
- Antivirus / EDR
- Next-generation firewall
- VPN
- RBAC permissions

- Email filtering sandbox
- Encryption
- SIEM monitoring
- Cloud backups

These directly defend systems from cyber threats.

Why These Recommendations Protect the Network

Linkchain Gaming stores valuable unannounced game IP, making it a target for insiders and external attackers. The recent credential-sharing incident proves weaknesses in identity management and access control.

By implementing MFA, RBAC, monitoring, patching, backups and staff training, the company significantly reduces:

- Insider threats
- Phishing compromise
- Malware infections
- Data theft
- Downtime
- Financial losses
- Reputational damage

Risk Summary (RAG)

Rating	Number of Risks
High	12
Medium	7
Low	1

The current environment presents a high cyber security risk level, requiring urgent remediation.

Conclusion

Linkchain Gaming's biggest risks are weak identity controls, phishing exposure, poor access permissions and insufficient resilience for remote working. If these

recommendations are implemented, the organisation will greatly improve confidentiality, integrity and availability of systems while protecting valuable intellectual property.

Assignment 3 – Cyber Security
Linkchain Gaming
Task 2 – Information Security Guidelines Recommendations

1. Introduction

Linkchain Gaming is a medium-sized UK game development company that has recently experienced a cyber security incident involving a disgruntled employee who shared their username and password. This resulted in unauthorised access to confidential information stored within shared folders, including unannounced intellectual property.

As the company continues to grow and modernise its enterprise network, it is essential that a robust Information Security Policy is introduced. The current controls are insufficient to protect company data, remote workers, systems and reputation.

This report recommends a range of security controls, explains how they should be enforced, and identifies relevant legislation, frameworks and standards.

2. Purpose of an Information Security Policy

The purpose of an Information Security Policy is to define how Linkchain Gaming protects:

- Confidential information
- Game source code
- Intellectual property
- Customer and employee data
- IT systems and cloud services
- Physical devices and premises

It also ensures all employees understand their responsibilities when handling information and using company systems.

3. Key Security Objectives

The policy should be designed around the CIA Triad:

Confidentiality

Only authorised users can access sensitive information.

Integrity

Data remains accurate, complete and unaltered.

Availability

Systems and data remain accessible when needed.

4. Recommended Controls

4.1 Access Control Policy

Recommendation

Linkchain Gaming should implement Role-Based Access Control (RBAC) so employees only access systems relevant to their job role.

Justification

The previous incident involved access to shared folders containing confidential intellectual property. Excessive permissions increase insider threat risk.

Enforcement

- Permissions linked to job role
- Manager approval required for access requests
- Quarterly access reviews
- Automatic removal of leavers' accounts

Relevant Standards

- ISO 27001 – Access Control
- GDPR UK Data Protection Principles

4.2 Password Policy

Recommendation

Strong password controls should be mandatory.

Requirements

- Minimum 14 characters
- Complexity rules
- Password manager encouraged
- No password sharing
- Forced reset after suspected compromise

Enforcement

- Active Directory password policy
- Account lockout after failed attempts

Justification

The company already suffered from shared credentials. This control directly addresses that weakness.

4.3 Multi-Factor Authentication (MFA)

Recommendation

MFA must be enabled for:

- Email
- VPN
- Cloud storage
- Admin accounts
- Source code repositories

Enforcement

Use Microsoft Authenticator or hardware tokens.

Justification

Even if passwords are stolen, attackers cannot access accounts without second-factor approval.

4.4 Acceptable Use Policy

Recommendation

Employees must understand acceptable behaviour when using company devices and systems.

Include:

- No unauthorised software
- No personal file sharing
- No suspicious downloads
- No password sharing
- Safe browsing expectations

Enforcement

Signed policy acknowledgement during onboarding.

4.5 Email Security Policy

Recommendation

Because Jordan opened all attachments automatically, email security must be strengthened.

Controls

- Attachment sandbox scanning
- Anti-phishing filters
- Block executable attachments
- Banner for external emails
- Report phishing button

Enforcement

Mail gateway filtering and user training.

Justification

Email remains one of the biggest attack vectors.

4.6 Remote Working Policy

Recommendation

As Jordan works from home full time, remote working controls are essential.

Controls

- Company-managed laptops only

- VPN required
- MFA required
- Home router guidance
- Screen lock after inactivity
- No family access to device

Enforcement

Endpoint management platform (Intune / MDM)

Justification

Remote staff create expanded attack surface outside office perimeter.

4.7 Endpoint Security Policy

Recommendation

All laptops/desktops should include:

- Antivirus / EDR
- Disk encryption
- Patch management
- USB control
- Device monitoring

Enforcement

Central security dashboard.

Justification

Protects against malware, theft and unauthorised software.

4.8 Data Protection Policy

Recommendation

Sensitive data should be classified:

Level	Example
Public	Marketing material

Internal	Staff procedures
Confidential	HR records
Highly Confidential	Source code, unreleased games

Controls

- Encryption at rest
- Encryption in transit
- Restricted sharing
- Data Loss Prevention (DLP)

Relevant Legislation

- UK GDPR
- Data Protection Act 2018

4.9 Backup and Recovery Policy

Recommendation

Use 3-2-1 backup model:

- 3 copies of data
- 2 different media types
- 1 offsite/cloud copy

Enforcement

Daily automated backups with monthly restore testing.

Justification

Protects against ransomware, deletion and outages.

4.10 Incident Response Policy

Recommendation

Formal process for detecting and responding to incidents.

Stages

1. Identification
2. Containment

3. Eradication
4. Recovery
5. Lessons learned

Enforcement

Incident manager assigned. All incidents logged.

Justification

Faster response reduces damage and downtime.

4.11 Joiners Movers Leavers Policy
Recommendation

HR and IT must coordinate account lifecycle management.

Controls

- Create accounts only after approval
- Amend permissions when role changes
- Disable accounts immediately on leaving
- Recover assets

Justification

Prevents former staff retaining access.

4.12 Security Awareness Training
Recommendation

Mandatory annual training covering:

- Phishing
- Password hygiene
- Data handling
- Social engineering
- Remote work security

Enforcement

Completion tracked by HR system.

Justification

Humans are commonly targeted.

5. Compliance and Legal Requirements
UK GDPR / Data Protection Act 2018

Protect personal data lawfully and securely.

Computer Misuse Act 1990

Unauthorised access or misuse of systems is illegal.

Copyright / Intellectual Property Law

Protect game code, media assets and unreleased titles.

Health and Safety at Work Act

Applies to office equipment, secure working environments and remote setups.

6. Recommended Frameworks and Standards
ISO 27001

Cyber Essentials Plus

Useful UK certification for defending against common attacks.

NIST Cybersecurity Framework

Good for identify, protect, detect, respond and recover model.

7. Policy Review and Expansion

As Linkchain Gaming grows, policies must scale.

Recommended Review Schedule

- Minor review every 6 months
- Full annual review
- Immediate review after incident
- Review after mergers/new offices/new cloud systems

Why Important

Rapid growth often creates unmanaged accounts, shadow IT and inconsistent controls.

8. Summary of Immediate Priorities

1. Implement MFA
2. Introduce RBAC
3. Tighten shared folder permissions
4. Deploy email filtering
5. Security awareness training
6. Endpoint management
7. Backup strategy
8. Incident response plan
9. Remote worker controls
10. ISO 27001 roadmap

9. Conclusion

Linkchain Gaming currently faces significant security risks due to weak credential management, excessive file access, email exposure and remote working vulnerabilities.

Implementing the controls in this report would create a stronger and more mature security posture. It would protect confidential intellectual property, reduce insider threats, improve resilience and support future business growth.

A formal Information Security Policy should now be approved by senior management and enforced across all departments.

Assignment 3 – Cyber Security
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Task 3 – Disaster Recovery Document

1. Introduction

I have been asked to recommend a range of actions that Linkchain Gaming can implement to provide effective disaster recovery support following a service outage caused by cyber incidents such as Denial of Service (DoS), Distributed Denial of Service (DDoS), ransomware, hardware failure or loss of connectivity.

Linkchain Gaming is a medium-sized UK game development company that stores confidential source code, intellectual property, internal communications and customer data. Because the business relies heavily on digital systems, prolonged downtime would create significant financial and reputational damage.

The company requires all critical services and business operations to be fully restored within **3 days** of a major disaster and has allocated an estimated budget of **£150,000** for this project.

This document recommends practical recovery measures, preventative controls and costed solutions.

2. Disaster Recovery Objectives

Primary Recovery Targets

Objective	Target
Maximum outage before recovery starts	Immediate
Critical systems restored	24 hours
Core business fully operational	72 hours
Maximum data loss (RPO)	4 hours
Full network recovery	Within 3 days

3. Main Causes of Service Outages

Cyber Causes

- DoS / DDoS attacks
- Ransomware
- Malware infection
- Phishing compromise
- Insider sabotage
- Credential theft

Technical Causes

- Server failure
- Internet outage
- Firewall failure
- Switch failure

- Power outage
- Corrupt updates

Physical Causes

- Fire
- Flood
- Theft
- Building access breach
- Environmental damage

4. Recommended Disaster Recovery Measures

4.1 Internet Resilience & DoS Protection

Linkchain Gaming relies on internet access for cloud systems, collaboration, downloads and remote workers.

Recommendations
Dual ISP Connectivity

Use two separate internet providers with automatic failover.

Example:

- BT Business Fibre
- Virgin Media Business

DDoS Protection Service

Use cloud scrubbing services to absorb attack traffic.

Firewall High Availability Pair

Two firewalls configured in failover mode.

Estimated Cost

Measure	Cost
Second ISP line	£4,000 per year
DDoS protection service	£12,000 per year

2 x Next Gen Firewalls	£18,000
Installation/configuration	£4,000

Subtotal: £38,000

4.2 Server Recovery & Virtualisation

Current businesses risk major disruption if servers fail.

Recommendations
Two Virtual Hosts

Install two physical servers running Hyper-V or VMware.

Replication Between Hosts

Virtual machines copied live between systems.

Automatic Failover

If one host fails, services restart on second host.

Estimated Cost

Measure	Cost
2 x Enterprise Servers	£16,000
Shared storage/NAS	£8,000
UPS power protection	£4,000
Setup & migration	£6,000

Subtotal: £34,000

4.3 Backup & Ransomware Recovery

Ransomware is one of the most serious outage threats.

Recommended Backup Strategy

Use 3-2-1 model:

- 3 copies of data
- 2 storage types

- 1 offsite immutable copy

Include:

- Daily cloud backup
- Hourly snapshot backups for critical data
- Monthly offline archive
- Quarterly restore testing

Estimated Cost

Measure	Cost
Cloud backup platform	£8,000 per year
Backup appliance	£7,000
Immutable storage add-on	£3,000
Testing and setup	£2,000

Subtotal: £20,000

4.4 Endpoint Recovery for Staff

Remote workers must continue working during outages.

Recommendations

- Microsoft 365 cloud productivity suite
- OneDrive autosave/version history
- Spare laptops ready for deployment
- Intune device management
- Remote wipe capability

Estimated Cost

Measure	Cost
10 spare laptops	£9,000
Intune / MDM licences	£4,000
Configuration labour	£2,000

Subtotal: £15,000

4.5 Office Power & Environmental Recovery

Recommendations

- UPS for servers/network
- Air conditioning in server room
- Fire suppression sensors
- CCTV and access control

Estimated Cost

Measure	Cost
UPS systems	£6,000
Cooling improvements	£5,000
Access control/CCTV	£7,000

Subtotal: £18,000

4.6 Incident Response & Business Continuity Planning

Technology alone is not enough.

Recommendations

- Written Disaster Recovery Plan
- Contact lists
- Escalation matrix
- Incident response team
- Tabletop exercises twice yearly
- Staff awareness training

Estimated Cost

Measure	Cost
Consultancy & DR documentation	£7,000
Staff exercises/training	£3,000

Subtotal: £10,000

5. Total Project Cost

Area	Cost
Internet & DoS Protection	£38,000
Server Recovery	£34,000
Backup & Ransomware	£20,000
Endpoint Recovery	£15,000
Physical Environment	£18,000
Planning & Training	£10,000

Total = £135,000

This remains within the available **£150,000 budget**, leaving **£15,000 contingency reserve**.

6. How the Plan Meets the 3-Day Recovery Target

Time After Incident	Recovery Action
0–1 Hour	Incident declared, team activated
1–4 Hours	Failover internet/firewall systems
4–12 Hours	Restore critical cloud access, communications
12–24 Hours	Virtual servers recovered
24–48 Hours	User devices reissued / reconnect remote staff
48–72 Hours	Full business operations restored

7. DoS / DDoS Recovery Procedure
If Attack Occurs

1. Detect abnormal traffic spikes
2. Activate ISP and DDoS provider response
3. Divert traffic to scrubbing centre
4. Geo-block malicious regions if needed
5. Enable rate limiting / WAF rules
6. Move users to secondary line
7. Communicate status to staff/customers

Why Effective

This prevents public services becoming unavailable and reduces downtime.

8. Ransomware Recovery Procedure
If Ransomware Detected

1. Isolate infected devices immediately
2. Disable compromised accounts
3. Disconnect affected systems
4. Begin forensic review
5. Restore clean backups
6. Reset credentials
7. Patch root cause
8. Resume services safely

Important

Paying ransom is not recommended unless advised legally and operationally.

9. Additional Preventative Controls

To reduce disasters occurring in the first place:

- MFA for all users
- Patch management
- Email filtering
- RBAC permissions
- Endpoint Detection & Response
- Vulnerability scanning
- Security awareness training
- Penetration testing annually

10. Why These Recommendations Better Protect the Company

These measures improve:

Availability

Systems remain online during attacks or failures.

Integrity

Backups and monitoring prevent corrupted systems from spreading.

Confidentiality

MFA, access control and encryption reduce unauthorised access.

Reputation

Faster recovery reduces customer trust damage.

Financial Protection

Less downtime means reduced lost revenue.

11. Final Recommendation Summary

Linkchain Gaming should immediately prioritise:

1. Dual internet with DDoS mitigation
2. Virtual server failover platform
3. Secure cloud backups
4. Device management for remote staff
5. Tested disaster recovery procedures
6. Staff training and exercises

12. Conclusion

Linkchain Gaming is highly dependent on digital systems and would face major disruption if impacted by a cyber incident or technical outage. This disaster recovery plan provides a realistic and affordable solution within the £150,000 budget.

The proposed controls would allow the business to restore critical services rapidly and return to full operation within the required **3-day target**, while also improving long-term resilience against ransomware, DoS attacks and infrastructure failure.